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EEL 3135 – Classwork #06

Milestone:

- Complete any 2 questions to receive credit.

Question #1:

Given:

$$x(t) = 8e^{j(6t-\pi/3)} + 8e^{-j(6t-\pi/3)}$$

(a) Use Euler's identity to convert $x(t)$ into its real sinusoidal form.

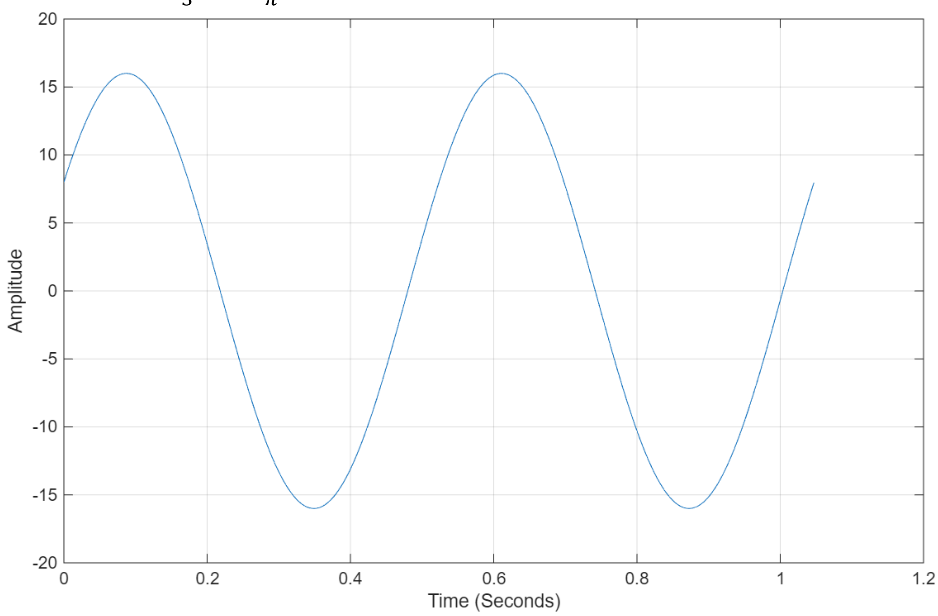
(b) Identify the amplitude, phase shift, and frequency of the resulting cosine wave.

(c) Generate and label a MATLAB plot showing two full periods of $x(t)$.

a) $x(t) = 16\cos\left(\frac{6}{\pi}t - \frac{\pi}{3}\right)$

b) A = 16; $\phi = -\frac{\pi}{3}$; $f = \frac{6}{\pi} = 1.90986\ Hz$

c)



Question #2: You are given the following frequency-domain information for $y(t)$:

$$\omega_0 = 3 \text{ rad/s}, \quad a_0 = 2, \quad a_{\pm 1} = 1.5e^{\mp j\pi/2}, \quad a_{\pm 2} = 0.5e^{\mp j\pi/4}$$

(a) Write $y(t)$ as a sum of complex exponentials.

(b) Rewrite $y(t)$ using cosine terms (real Fourier series).

(c) Plot $y(t)$ over two fundamental periods in MATLAB with proper axis labels.

Question #3: Consider the signal

$$x(t) = 10\cos(400\pi t) + 4\cos(700\pi t)$$

(a) Determine the highest frequency component present in $x(t)$.
- $4\cos(700\pi t) \rightarrow \omega = 700\pi\ ;\ f = 350\ Hz.$

(b) Find the minimum sampling frequency (Nyquist rate) required to avoid aliasing.

$$f_s \geq 2f_{max} \rightarrow f_s \geq 700\ Hz$$

(c) If the sampling rate is f_s = 500 Hz, compute the aliased frequency (if any) for each cosine term and determine the resulting discrete-time frequencies.

Term 1: $f < f_s \rightarrow no\ aliasing \rightarrow 200\ Hz$

Term 2: $1 * \frac{f_s}{2} - 500\ Hz = 350\ Hz - 500\ Hz = 150\ Hz$

$$w_{d1} = 2\pi * \frac{f_a}{f_s} = 2\pi * \frac{200}{500} = 0.8\ radians\ per\ sample$$

$$w_{d2} = 2\pi * \frac{f_a}{f_s} = 2\pi * \frac{150}{500} = 0.6\ radians\ per\ sample$$

Question #4: Consider the continuous-time signal

$$x(t) = 6\cos(150\pi t) - 4\cos(250\pi t)$$

(a) Determine the highest frequency component in $x(t)$.

(b) Compute the Nyquist rate.

(c) If the signal is sampled at $f_s = 180$ Hz, state whether aliasing will occur and explain why.